Lab 4: Temporal Difference Learning

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Section 1: Project Overview

This lab builds upon Monte Carlo (MC) and Dynamic Programming (DP) by introducing Temporal Difference (TD) Learning. TD operates on-policy, meaning it updates every step instead of at the end of each episode. It does so by using bootstrapping, which involves updating value estimates without a complete model, which leads to faster training/learning. This lab uses the same agent structure while only differing by one line of code: a SARSA agent uses on-policy, and a Q-learning agent updates off-policy. This alteration leads to two very different approaches in CliffWalking.

The environment used in the lab follows:

CliffWalking-v1 (Gymnasium):

- State space: 48 discrete states (4 rows $\times$ 12 columns), integer 0–47
- Action space: 4 discrete actions — up, right, down, left
- Rewards: -1 per step, -100 for cliff (resets to start)
- Terminal condition: reaching state 47 (bottom-right goal)

Since SARSA is updated based on the action it takes, I estimate that it will follow the least risky path in the CliffWalking environment. The Q-learning agent however will take much higher risk to find the most optimal solution. I anticipate training to be more negative for the Q-learning agent but to have the more optimal policy after training is complete.

Section 2: Deliverables

GitHub Repository

`GitHub Repository: https://github.com/cooper-rm/temporal-difference-learning`

Implementation Summary

I implemented SARSA and Q-learning on CliffWalking-v1. For each of the agents I used an epsilon greedy policy with epsilon=0.1, alpha=0.5, gamma=1.0, and a Q-table from a numpy array. For each I ran 30 independent experiments based on 30 different seeds. Each experiment was run over 1000 episodes and plotted for visual inspection. Additionally, I experimented with different hyperparameters varying alpha (0.1, 0.3, 0.5) and epsilon-decay (constant, linear, exponential).

Key Results & Analysis

SARSA averaged -28.0 reward over last 100 episodes, whereas Q-learning averaged -51.6 as seen in Figure 1. SARSA was 23.5 points better during training because it avoids the cliff. On the other hand, Q-learning falls off the cliff more during training due to exploration along the cliff's edge.

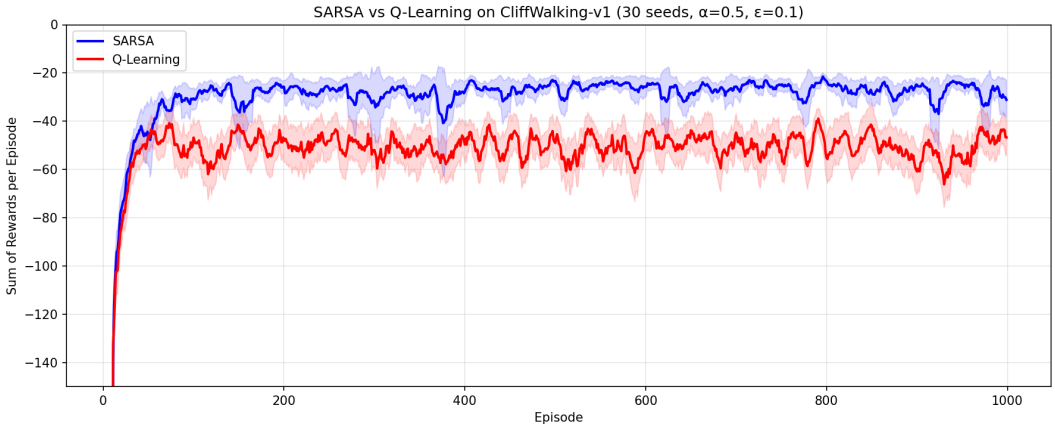

Figure 1: SARSA vs Q-learning learning curves averaged over 30 seeds with 95% confidence intervals. Sum of rewards per episode over 1000 training episodes with $\alpha = 0.5$ and $\epsilon = 0.1$.

When we look at the learned policies for SARSA and Q-learning we can see very different paths (Figure 2 & Figure 3). SARSA arrows go up from start, right across the top row, then down to goal. SARSA takes the safest path of 17 steps and scores a reward of -17 to stay as far as possible from the cliff. Q-learning arrows go right along row 2, directly above the cliff edge, then down to goal. This is the optimal path, taking only 13 steps and scoring a reward of -13.

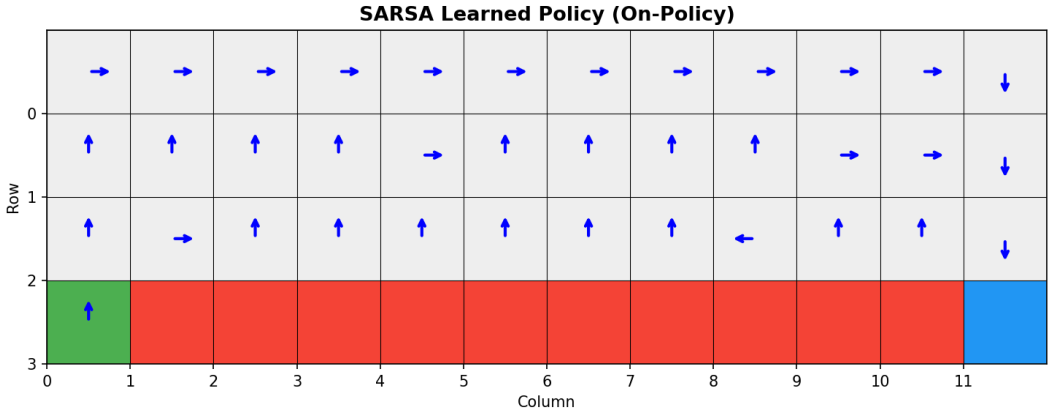

Figure 2: SARSA greedy policy arrows showing the learned action at each grid state after 2000 episodes of training. Green = start, blue = goal, red = cliff.

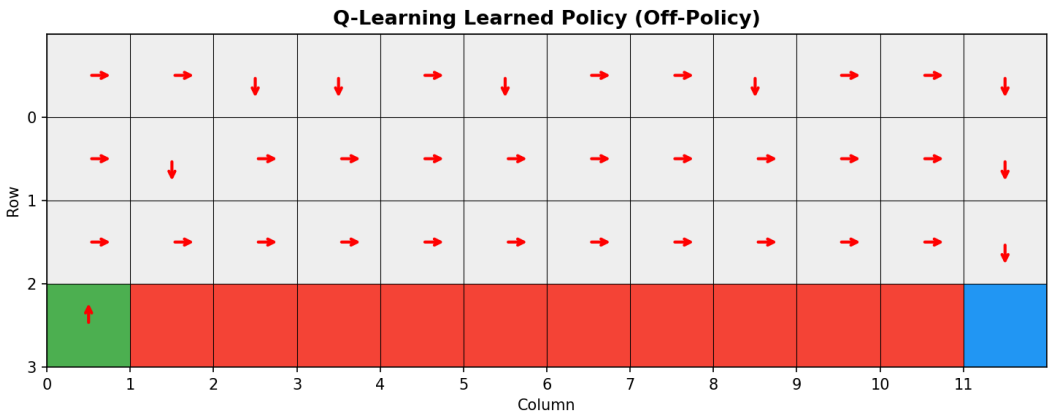

Figure 3: Q-learning greedy policy arrows showing the learned action at each grid state after 2000 episodes of training. Green = start, blue = goal, red = cliff.

When experimenting with differing epsilon decay schedules (Figure 4), exponential decay improved Q-learning the most, changing from -50.5 to -16.8. This happens because epsilon drops close to zero quickly and the agent stops falling off the cliff as much. Linear epsilon decay still improved the policy over time but much slower. Finally, exponential decay seemed to work the best given enough training time. With enough time, Q-learning eventually beats SARSA because the optimal cliff path becomes safe to follow after epsilon has decayed enough.

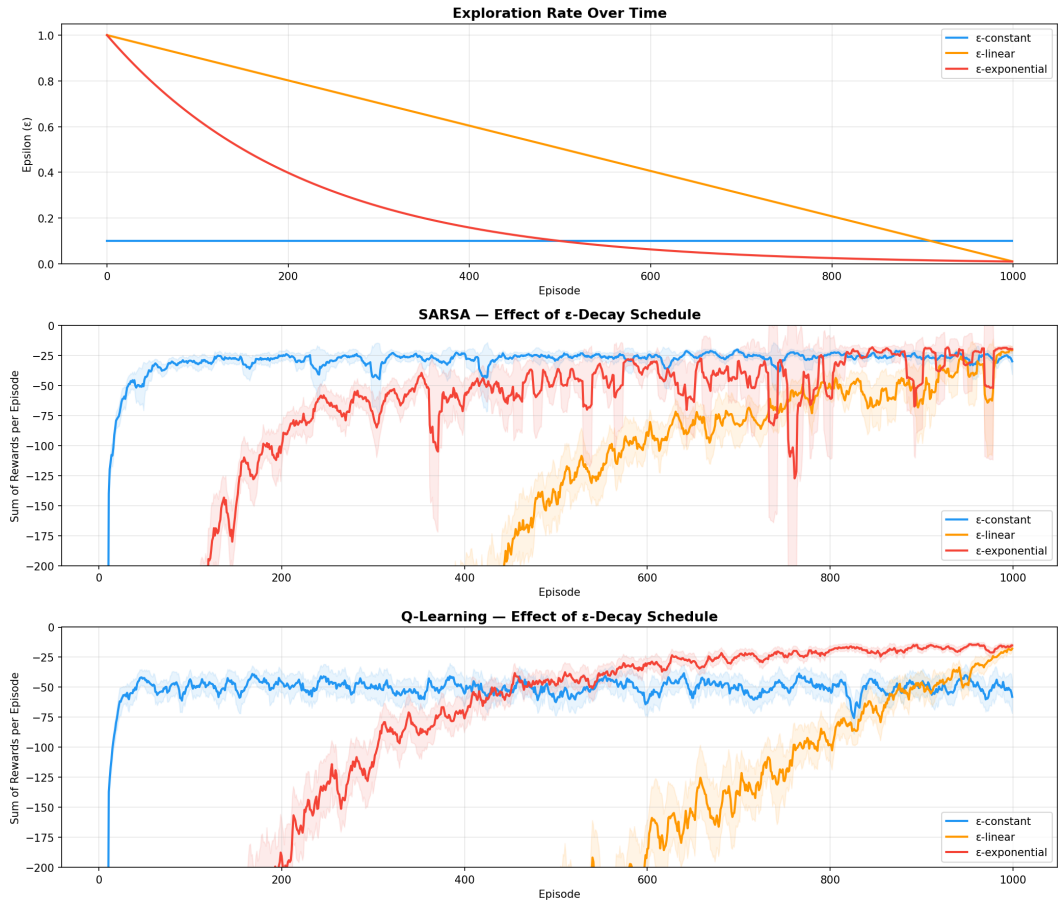

Figure 4: Epsilon decay schedule comparison. Top: epsilon value over time for each schedule. Middle: SARSA reward curves. Bottom: Q-learning reward curves. All averaged over 20 seeds.

Section 3: AI Use Reflection

Initial Interaction

Like previous labs, I used ChatGPT to first discuss temporal difference learning. I had it review the concepts in the week's materials and then interview me. Once I felt confident in this week's concepts, I switched to Claude Code where I began working through Lab 4. Previous weeks felt very easy, but this week Claude made a number of silly mistakes, as discussed in the next few sections.

Iteration Cycle

Iteration 1: Notebook Built Backwards

Initially, I had Claude discuss a plan of action and begin with templating Lab 4 as a Jupyter notebook. After discussing the details, it wrote the notebook with each section in a markdown cell, however it completed the sections in the wrong order starting at part 7 and

counting back to part 1. This was interesting, since I had not encountered issues like this with Claude Code in the past. After some discussion, the order was corrected and I was able to move on.

Iteration 2: Deprecated Environment Version

The next issue I found was that CliffWalking-v0 threw a deprecation warning. I discussed it with Claude, and all cells relating to CliffWalking-v0 were updated to CliffWalking-v1. This was a quick and simple fix, but made it clear that Claude doesn't always understand the latest API versions.

Iteration 3: SARSA Greedy Rollout Loop Bug

When working on the trajectory plot for SARSA, the plot showed SARSA going 1 step then looping back to start. This was of course incorrect since the training was correctly working. After discussing this issue with Claude, it identified the root cause to be that the numpy random state had drifted after running Parts 2 through 5, so the Q-table was different. This was not an easy fix, as it took multiple attempts to figure out.

Critical Evaluation

Claude made more small errors this lab than all other previous labs. This included organizational errors, logic and code errors, and issues with linting. Most of these were caught by myself reading through the code or running each cell, and then working with Claude to fix them. Fortunately, I was able to verify each step and come to a reasonable code base in the end, matching up with expectations from Sutton and Barto.

Learning Reflection

The trajectory bug taught me that TD methods are sensitive to random state and need enough training to find the final correct policy. I was also able to understand that SARSA and Q-learning are the same agent, with a single different line of code for updates being the difference. Additionally, working with AI still requires careful review, as seen via Claude's small but consistent errors during this lab.

Section 4: Speaker Notes

- **Problem:** Explore temporal difference learning on CliffWalking.
- **Method:** SARSA and Q-learning with epsilon-greedy, 30 seeds.
- **Design choice:** Same agent, one line different in the update.
- **Key result:** SARSA takes the safe path (-28.0), Q-learning takes the cliff edge (-51.6) but learns the optimal policy (-13).
- **Insight:** Epsilon decay closes the gap between the two algorithms.
- **Challenge:** SARSA trajectory bugged out due to random state drift, took a few tries to fix.

- **Connection:** TD bridges Monte Carlo and Dynamic Programming, learning step by step without a model.

References

1. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.
2. Anthropic. (2025). Claude Code [Large language model CLI tool]. `https://claude.ai`
3. OpenAI. (2025). ChatGPT [Large language model]. `https://chat.openai.com`